EE3014 ENERGY CONVERSION LAB

WEEK1

1) As shown in Figure 1, a permanent magnet (PM) composed of a neodymium-iron-boron alloy is placed in the magnetic circuit.

$A_g = 4\ \text{cm}^2$, $l_g = 0.4\ \text{cm}$

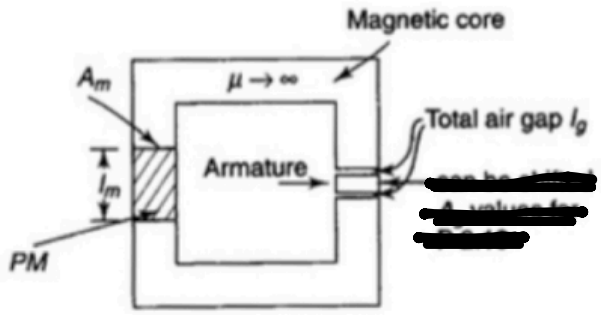
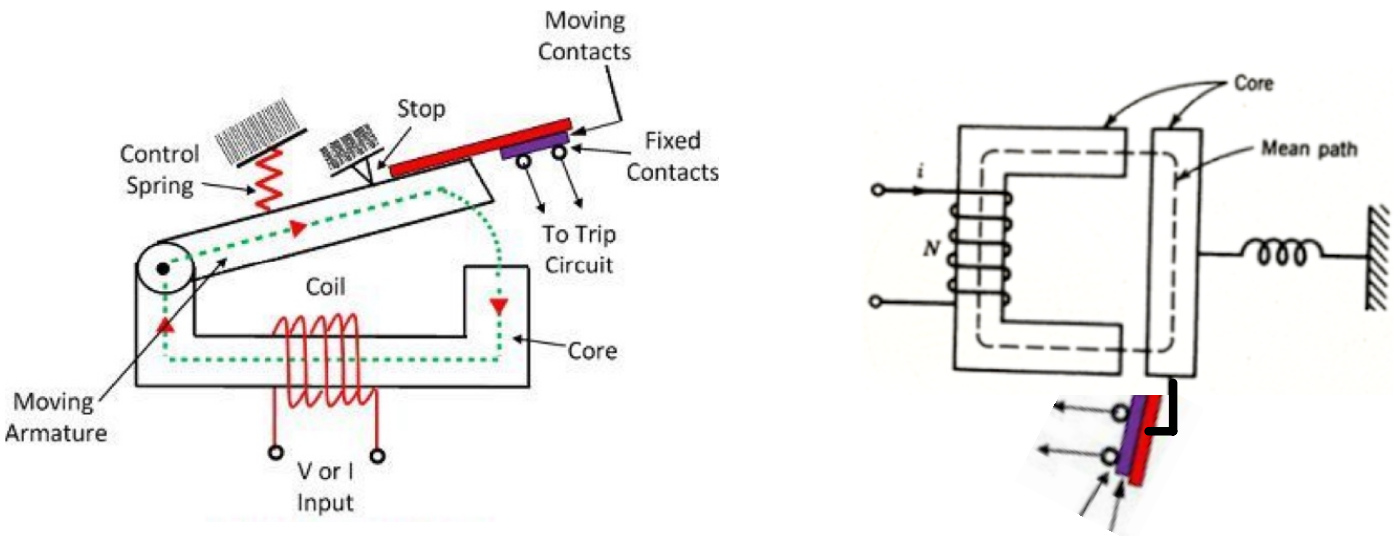


Figure 1 Magnetic circuit

a) It is required that the air gap flux density Bg will be 0.5 T. Determine l_m to get the optimum design (the minimum volume of PM).
b) Also, write the Matlab source script to get the same solution.

2) Figure 2 represent the magnetic circuits of relays The coils have 500 turns each and the mean core paths are lc = 325 mm (each). When the air gap lengths are 1.5mm each, a flux density of 0.75 Tesla is required to actuate the relay. The core is cast iron, as shown in Figure 3.

a) Find the currents in the coils.
b) Compute the values of permeability and relative permeabilty of the core.
c) If the air gap is zero, find the current in the coil for the same flux density (0.75 T) in the core.
d) Repeat the solutions of a), b), and c) for the cast steel and comment on the results.



a) b)

Figure 2. a) Hinged and b) primitive type of armature relays

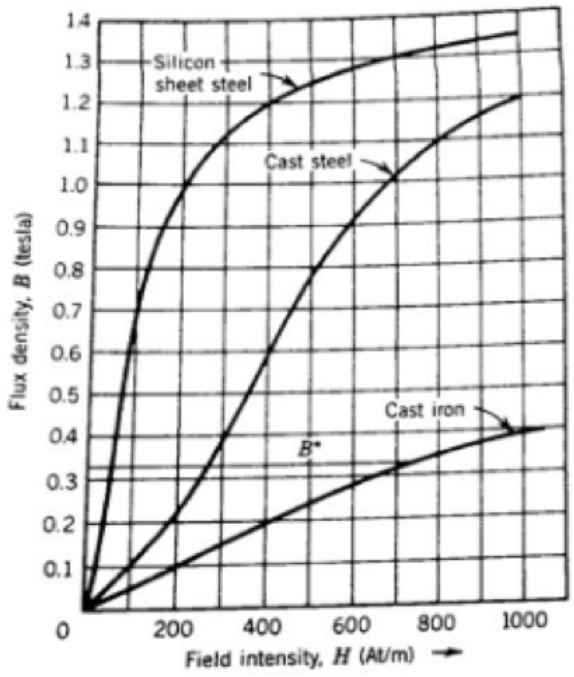


Figure 3. Magnetic curves of different materials